Q1- The incorrect statement of the following is:

A- Unused chemicals shouldn't be returned to the stock bottles.
B- When dealing with flammable liquid heat in water bath.
C- You can carry reagent bottles to your desk and take a sample.
D- Always prepare your experiment before take it in Lab.

Q2- The incorrect statement of the following is:

A- To avoid chemical loss, unused chemicals should be returned to the stock bottles.
B- When dealing with flammable liquid heat in blue flam directly.
C- You can't carry reagent bottles to your desk and take a sample.
D- Always discard waste according to your instructor order.

Q3- The correct statement of the following is:

A- To avoid chemical loss, unused chemicals should be returned to the stock bottles.
B- You can heat all types of liquid in blue flam directly.
C- You can carry reagent bottles to your desk and take a sample.
D- Long hair must be tied back all lab period.

Q4- The correct statement of the following is:

A. All chemicals are hazardous.
B. You can do your lab experiment without wearing lab coat.
C. Eating, drinking and phoning are allowed in the lab.
D. To avoid chemical loss, unused chemicals should be returned to the stock bottles.

Q5- The beside pictogram means that this material is:



A. Toxic **B. Corrosive** **C. Harmful** **D. Flammable**

Q6- The beside pictogram means that this material is:



A. Toxic **B. Corrosive** **C. Harmful** **D. Explosive**

Q7- The beside pictogram means that this material is:



A. Toxic **B. Corrosive** **C. Harmful** **D. Flammable**

Q8- The beside pictogram means that this material is:



A. Toxic **B. Corrosive** **C. Harmful** **D. Flammable**

Q9- Which type(s) of molecule(s) are polar solvents more likely to be able to dissolve?

A. ionic molecules B. polar and ionic molecules
C. polar molecules D. ionic, polar and non-polar molecules

Q10- Which of the following will dissolve in water to produce a clear solution?
A. $CaCl_2$ B. NaOH C. $CH_3(CH_2)_8OH$ D. AgCl

Q11- Which of the following will dissolve in water to produce a clear solution?
A. $CaCl_2$ B. KBr C. $CH_3(CH_2)_8OH$ D. AgBr

Q12- Which one of the following will dissolve in Hexane :

A. I_2 B. H_2O C. NaCl D. CH_3OH

Q13- Which one of the following will dissolve in Hexane :

A. C_6H_6 B.NaCl C.H_2O D.KBr

Q14- Which equipment is used to measure the volume of a solid in density determination experiment:
A. Volumetric cylinder B- Balance
C- Thermometer D- Graduated cylinder

Q15- Which equipment is used to indicate the boiling point for a liquid:
A- Capillary tube B- Balance C- Thermometer D- Test tube

Q16- Which equipment is used to measure the mass of a liquid in density experiment :
A- Capillary tube B- Balance C- Thermometer D- Test tube

Q17- The type of heating that you use in the boiling point determination is:
A) Heat over flame directly B) Water bath C) Ethanol bath D) Benzene bath

Q18- During density determination experiment a student collect the following set of data for a solid:

Mass of solid	5.7g
Initial volume of water	12.4mL
Final volume of water	17.0mL

Then the density of a solid =:
A- 1.47g/cm^3 B-1.23 g/ cm^3
C- 1.09 g/ cm^3 D-2.60 g/ cm^3

Q19- During density determination experiment a student collect the following set of data for a solid:

Mass of solid	15.7g
Initial volume of water	12.5mL
Final volume of water	14.0mL

Then the density of a solid =:
A- 0.75 g/cm^3 B-0.23 g/ cm^3
C- 1.09 g/ cm^3 D-0.51 g/ cm^3

Q20- During density determination experiment a student collect the following set of data for a solid:
Then the density of a solid =:

Mass of solid	7.5g
Initial volume of water	12.5mL
Final volume of water	17.6mL

A- 1.47g/cm^3 **B-1.23 g/ cm^3**
C- 1.09 g/ cm^3 **D-2.60 g/ cm^3**

Q21- During density determination experiment a student collect the following set of data for a solid:
Then the density of a solid =:

Mass of solid	11.7g
Initial volume of water	12.5mL
Final volume of water	17.0mL

A- 1.47g/cm^3 **B-1.23 g/ cm^3**
C- 1.09 g/ cm^3 **D-2.60 g/ cm^3**

Q22- Which of the following pressure will give you the highest boiling point of Acetone:
A- 1.5 atm **B- 1 atm** **C- 1.85 atm** **D- 0.52 atm**

Q23- Which of the following pressure will give you the lowest boiling point of Acetone:
A- 1.5 atm **B- 0.95 atm** **C- 1 atm** **D- 0.52 atm**

Q24- A Chrome alum ($KCr(SO_4)_2.12H_2O$),(Mm alum =499g/mol), Calculate the mass of water in 12.9 g sample of this alum:
A- 7.02 g **B- 5.58 g** **C- 2.92 g** **D- 3.98 g**

Q25- A Chrome alum ($KCr(SO_4)_2.12H_2O$),(Mm alum =499g/mol), Calculate the mass of water in 18.9 g sample of this alum:
A- 7.02 g **B- 8.18 g** **C- 2.92 g** **D- 3.98 g**

Q26- A Chrome alum ($KCr(SO_4)_2.12H_2O$),(Mm alum =499g/mol), Calculate the mass of water in 6.75 g sample of this alum:
A- 7.02 g **B- 5.58 g** **C- 2.92 g** **D- 3.98 g**

Q27- A Chrome alum ($KCr(SO_4)_2.12H_2O$),(Mm alum =499g/mol), Calculate the mass of water in 15.6 g sample of this alum:
A- 6.75 g **B- 5.58 g** **C- 2.92 g** **D- 3.98 g**

Q28- Which one of these salts is an Alum:
A) $CuSO_4.5H_2O$ **B) $NaAl(SO_3)_2.12H_2O$** **C) $KAl(SO_4)_2.12H_2O$** **D) $CaMg(SO_4)_2.10H_2O$**

Q29- Which one of these salts is an Alum:

A) $CuSO_4.5H_2O$ B) $NaAl(SO_4)_2.12H_2O$ C) $KAl(SO_3)_2.12H_2O$ D) $CaMg(SO_4)_2.10H_2O$

Q30- If the % mass of H_2O = 37.5%, and the molar mass (Mm) of anhydrous salt = 379 g/mol. The value of water of crystallization (X) is:

A) 13 B) 12 C) 16 D) 10

Q31- If the % mass of H_2O = 42.5%, and the molar mass (Mm) of anhydrous salt = 379 g/mol. The value of water of crystallization (X) is:

A) 13 B) 12 C) 16 D) 10

Q32- If the % mass of H_2O = 37.5%, and the molar mass (Mm) of anhydrous salt = 279 g/mol. The value of water of crystallization (X) is:

A) 13 B) 12 C) 16 D) 9

Q33- Calculate the molecular weight of 1.58g of a volatile liquid (condensed vapor) which occupied a volume of 685ml at 0.921 atm and 375K (R= 0.0821 atm.L.K^{-1}.mol^{-1}):

A- 95.6 g/mol B- 91.6 g/mol C- 112.3g/mol D- 77.10 g/mol

Q34- Calculate the molecular weight of 1.58g of a volatile liquid (condensed vapor) which occupied a volume of 685ml at 0.921 atm and 395K (R= 0.0821 atm.L.K^{-1}.mol^{-1}):

A- 95.6 g/mol B- 81.2 g/mol C- 112.3g/mol D- 77.10 g/mol

Q35- Calculate the molecular weight of 1.85g of a volatile liquid (condensed vapor) which occupied a volume of 0.685 L at 0.932 mmHg and 88c° (R= 0.0821 atm.L.K^{-1}.mol^{-1}):

A- 95.6 g/mol B- 91.6 g/mol C- 112.3g/mol D- 85.9 g/mol

Q36 - Calculate the molecular weight of 1.85g of a volatile liquid (condensed vapor) which occupied a volume of 0.685 L at 0.932 mmHg and 79c° (R= 0.0821 atm.L.K^{-1}.mol^{-1}):

A- 95.6 g/mol B- 91.6 g/mol C- 112.3g/mol D- 83.7g/mol

Q37- During the Empirical formula of Magnesium oxide experiment , you get rid of by product which is:

A- NH_3 B- Mg_2N_3 C- H_2O D- Mg_3N_2

Q38- During the Empirical formula of Magnesium oxide experiment, you get rid of by product by adding:

A- NH_3 B- Mg_2N_3 C- H_2O D- Mg_3N_2

Q39- During the Empirical formula of Magnesium oxide experiment, you get rid of by product and smell a gas which is :

A- NH_3 **B- Mg_2N_3** **C- H_2O** **D- Mg_3N_2**

Q40- The calculated ratio between Mg/O in your experiment will be ……, if you Forgetting to weigh the cover along with the crucible and contents at the end.

(a) Increase **(b) Decrease**

(c) Not affected **(d) Become double**

Q41- Which pair of molecular formula and empirical formula is CORRECT?

(a) As_2O_5 - AsO_3 **(b) SO_2 - S_2O_4** **(c) C_3H_6 - CH_3** **(d) $C_6H_{12}O_6$ - CH_2O**

Q42- A compound with an empirical formula of CH_2 has a molar mass of 56 g/mol, What is its molecular formula :

A- CH_2 **B- C_2H_4** **C- C_4H_8** **D- C_5H_{10}**

Q43- Analysis a 14. 40 g sample of liquid compound composed of carbon (C= 7.23 g / Mm=12) , Hydrogen (H=1.60 g/ Mm=1) and the remains is Nitrogen (N Mm=14), the empirical formula for this compound is :

A- C_3H_8N **B- C_2H_6N** **C-C_3H_9N** **D- $C_3H_8N_2$**

Q44- Analysis a 11.89 g sample of liquid compound composed of carbon (C= 7.35 g/ Mm=12) , Hydrogen (H=1.64 g/ Mm=1) and the remains is Nitrogen (N Mm=14), the empirical formula for this compound is :

A- C_3H_8N **B- C_2H_6N** **C-C_3H_9N** **D- $C_3H_8N_2$**

Q45- The burning of Mg metal given an indication is :

A) Expanded **B) White flash** **C) Frothed** **D) Turns color to blue**

Q46- The method that you used in lab to determine the empirical formula for Mg oxide is:

A) Decomposition method **B) Analysis method** **C) Synthesis method** **D) Dumas method**

Q47- An oxoacid contains Hydrogen (H) 0.18 g, Oxygen (O) 5.72 g and Chlorine (Cl) 6.90 g, then the empirical formula of acid is:

A) HClO **B) $HClO_2$** **C) $HClO_3$** **D) $HClO_4$**

Q48- An oxoacid contains Hydrogen (H) 0.17 g, Oxygen (O) 8.70 g and Chlorine (Cl) 6.85 g, then the empirical formula of acid is:

A) HClO **B) $HClO_2$** **C) $HClO_3$** **D) $HClO_4$**

Q49- How many grams of Oxygen (O_2) Mm=16 g/mol are there in a 50.0L gas cylinder at 21◦c when the pressure is 1.7 atm:

A) 112.7 g **B) 56.3 g** **C) 116.2 g** **D) 58.1 g**

Q50- How many grams of Oxygen (O_2) Mm = 16 g/mol ,are there in a 50.0L gas cylinder at 12◦c when the pressure is 1.7 atm:

A) 112.7 g **B) 56.3 g** **C) 116.2 g** **D) 58.1 g**

Q51 - Which of the following cases increase the calculated molecular weight of volatile liquid:

A) Using pressure of (1 atm) instead of the actual external pressure (738 mmHg) in lab .

B) The recording mass of empty flask higher than the actual value.

C) Using a volume which is written on flask instead of the volume taken by cylinder.

D) The reported temperature is 5◦c lower than actual one.

Q52- Which of the following cases increase the calculated molecular weight of volatile liquid:

A) Using pressure of (1 atm) instead of the actual external pressure (744 mmHg) in lab.

B) The recording mass of empty flask lower than the actual value.

C) Using a volume taken by cylinder not the one written on the flask.

D) The reported temperature is 5◦c lower than actual one.

